AtlasLegalAI: A Legal AI Chatbot for Moroccan Law

Using RAG and Fine-Tuned Large Language Models

Boukaidi Tarik[1], **BEL EL YAZID Ilyas**[2], **Daoudi Moaad**[3], **Chahba Brahim**[4], **Cherradi Mohammed**[5]

Digital Transformation and Artificial Intelligence

National School of Applied Sciences of Al Hoceima

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Abstract

AtlasLegalAI is a specialized legal chatbot designed to improve access to Moroccan legal information through the integration of large language models (LLMs), parameter-efficient fine-tuning (PEFT), and retrieval-augmented generation (RAG) techniques. The system is built using a curated corpus of digitized Moroccan legal codes collected from official government sources and focuses on providing accurate and context-aware legal assistance in both French and Arabic, reflecting the multilingual nature of Moroccan legal documentation. To achieve efficient domain adaptation under limited computational resources, the Qwen2.5-3B model was fine-tuned using Low-Rank Adaptation (LoRA) with the Unsloth framework on CUDA-enabled GPU infrastructure. In order to reduce hallucinations and improve factual reliability, the chatbot integrates a RAG pipeline based on FAISS and the multilingual-e5-large embedding model for semantic retrieval of relevant legal passages. AtlasLegalAI is deployed as a full-stack web application using React for the frontend and FastAPI for the backend, providing an accessible and interactive interface for end users. The proposed approach demonstrates the effectiveness of combining PEFT-based fine-tuning with retrieval-augmented architectures for domain-specific legal AI applications while offering a reproducible, scalable, and computationally efficient framework for multilingual legal assistance systems.

Index Terms — Legal Chatbot, Moroccan Law, Deep Learning, Large Language Models, LLM Fine-Tuning, Qwen2.5, LoRA, PEFT, RAG, FAISS, NLP, Bilingual AI, GPU Inference, Unsloth.

I. Introduction

Access to legal information remains a profound practical challenge in Morocco and in many multilingual legal environments. Although major statutory texts are publicly available, they are often distributed across heterogeneous portals, formal PDF documents, and language-specific repositories. Moroccan legal documentation is particularly demanding because it combines French and Arabic sources, civil-law codification, and a body of administrative practice that is difficult for non-specialists to search. The result is a persistent access gap: citizens may have formal access to the law while lacking the practical ability to identify the relevant article, interpret its scope, or understand how it applies to an everyday legal question.

Large language models (LLMs) offer a promising interface for this problem because they can translate formal legal language into accessible explanations and support natural questions in multiple languages. However, legal assistance is a high-stakes domain: an answer that is fluent but unsupported by the official legal text may mislead users. General-purpose LLMs are therefore insufficient when used alone, especially in jurisdictions that are under-represented in training data. AtlasLegalAI addresses this limitation by combining a compact fine-tuned multilingual model with retrieval from official Moroccan legal sources.

The main contributions of this work are the development of a curated bilingual Moroccan-law corpus prepared from official legal portals and serialized for both supervised fine-tuning and retrieval, together with a resource-efficient adaptation of Qwen2.5-3B using LoRA/QLoRA that can be trained on Colab-class CUDA hardware. This work also introduces a FAISS-based multilingual RAG pipeline that grounds legal answers in retrieved statutory passages and helps reduce unsupported citations. In addition, it proposes a local-first React/FastAPI chatbot architecture designed for bilingual interaction and privacy-preserving deployment. Finally, the study provides a reproducible benchmark protocol and an 18-question multilingual test set for comparing AtlasLegalAI against external zero-shot baselines.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the methodology. Section IV describes the web application architecture. Section V reports experimental results and benchmark status. Section VI discusses limitations, and Section VII concludes with future work.

II. Related Work

The application of artificial intelligence to legal information retrieval and question answering has attracted growing interest over the past decade. Early work such as DoNotPay demonstrated the feasibility of rule-based legal chatbots for simple consumer disputes, while ROSS Intelligence introduced the first commercial AI-powered legal research platform based on IBM Watson, enabling semantic search over large bodies of case law. These systems, however, were designed primarily for common-law jurisdictions and English-language corpora, leaving civil-law and multilingual legal systems largely underserved.

The introduction of transformer-based large language models (LLMs), particularly BERT and its multilingual variants, opened new avenues for Arabic and French legal NLP. AraBERT provided pre-trained representations for Modern Standard Arabic, enabling downstream tasks such as named-entity recognition and text classification on Arabic legal documents. Subsequent models such as CamemBERT extended coverage to French, though none was specifically fine-tuned for Moroccan statutory law written in Arabic and French.

Parameter-efficient fine-tuning (PEFT) methods have significantly lowered the barrier to domain adaptation of large generative models. Hu et al.proposed Low-Rank Adaptation (LoRA), which inserts trainable low-rank matrices into the attention layers of a frozen pre-trained model, reducing the number of trainable parameters by several orders of magnitude while preserving task performance. This technique has since been adopted widely for adapting LLMs to specialized domains including medicine, law, and finance, and forms the core fine-tuning strategy of AtlasLegalAI.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), addresses a fundamental limitation of purely parametric LLMs: their inability to reliably cite specific facts or legal article numbers without hallucinating. By coupling a dense retrieval module with a generative decoder, RAG systems ground each response in retrieved external documents, dramatically reducing hallucination rates in knowledge-intensive tasks. Subsequent work on dense passage retrieval, FAISS-based approximate nearest-neighbor indexing, and multilingual sentence encoders such as multilingual-E5 has made RAG practical for multilingual and low-resource legal corpora, directly motivating the architectural choices of AtlasLegalAI.

Despite these advances, no prior system specifically targets the Moroccan legal context, which is characterized by a mixed French–Arabic corpus, a civil-law tradition rooted in both Islamic jurisprudence and French codification, and severe resource constraints on both the user and infrastructure sides. AtlasLegalAI fills this gap by combining LoRA-based fine-tuning of a compact multilingual LLM with a FAISS-indexed RAG pipeline, purpose-built for Moroccan statutory law and deployable on consumer GPU hardware.

III. Methodology

AtlasLegalAI follows a four-stage pipeline: Data Engineering, Fine-Tuning, RAG Construction, and System Integration. Figure 1 gives the reader the overall map before the detailed sub-sections.

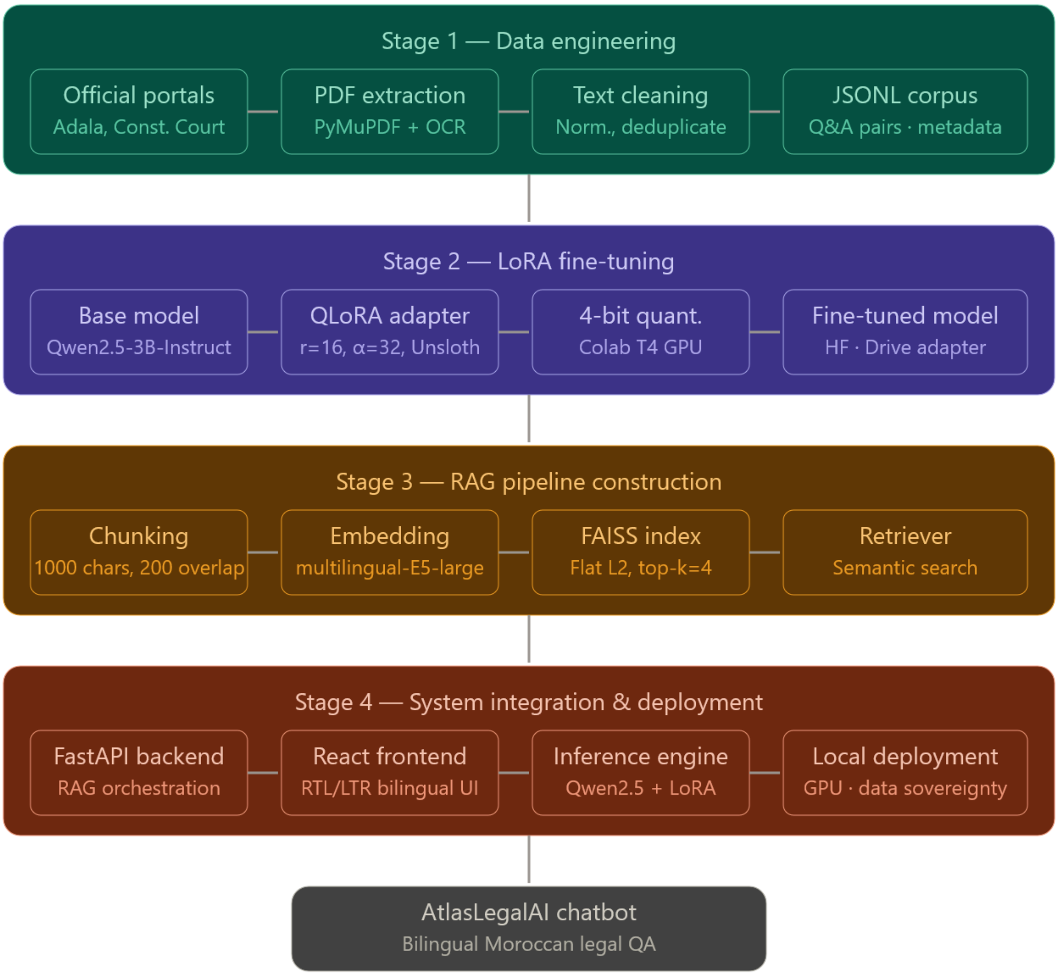


Fig. 1. Four-stage methodology pipeline of AtlasLegalAI

Table I. Technologies and tools used in the methodology.

Stage	Technology / Tool	Role in the Methodology
Data engineering	Official Moroccan legal portals	Primary authoritative sources for statutory and constitutional legal material.
Data engineering	PyMuPDF and Tesseract OCR	PDF text extraction, with OCR fallback for scanned French and Arabic documents.
Data engineering	JSONL	Structured serialization format for fine-tuning examples and retrieval records.
Fine-tuning	Qwen2.5-3B-Instruct	Multilingual generative backbone adapted to Moroccan legal question answering.
Fine-tuning	LoRA / QLoRA	Parameter-efficient adaptation of the base model under limited GPU memory.
Fine-tuning	Unsloth	Efficient 4-bit model loading, LoRA training, and inference optimization.
RAG construction	RecursiveCharacterTextSplitter	Segmentation of legal records into overlapping article-aware chunks.
RAG construction	intfloat/multilingual-e5-large	Dense multilingual embeddings for Arabic and French legal retrieval.
RAG construction	FAISS	Local vector index for fast semantic similarity search over legal chunks.
System integration	FastAPI and Uvicorn	Backend orchestration for retrieval, prompt construction, and model inference.
System integration	React, Vite, TypeScript, TailwindCSS	Responsive bilingual web interface with Arabic RTL and French LTR support.

A. Data Engineering

The corpus was collected from official Moroccan legal portals and converted into structured records suitable for both supervised fine-tuning and vector retrieval. Table I summarizes the primary sources.

Portal	Description & Scope
Adala Justice Portal adala.justice.gov.ma	Official digital repository of the Moroccan Ministry of Justice. Hosts the full corpus of Moroccan legal codes in French, including the Code de la Famille (Moudawana), the Penal Code, the Civil Procedure Code, the Commercial Code, the Labor Code, and numerous ministerial decrees. Documents are published as structured PDF files organized by legal domain.
Constitutional Court cour-constitutionnelle.ma	Official website of the Moroccan Constitutional Court. Provides access to constitutional rulings, decisions, and the consolidated text of the 2011 Moroccan Constitution. Enriches the corpus with constitutional jurisprudence, enabling the system to respond to queries at the intersection of ordinary law and constitutional principles.

2) Data Acquisition

- All PDF documents were manually downloaded from the official legal portals listed above. Each file was retrieved individually through the standard public web interfaces and stored locally. The acquisition process followed three main phases:
- Discovery: Each portal’s document index was manually explored to identify all available PDF resources. On the Adala portal, documents are organized under /resources/46, grouped by legal domain.
- Download: Each identified PDF was downloaded directly and stored in a structured local directory, organized by domain (e.g., *family_law/*, *penal_code/*, *commercial_law/*)
- Deduplication: Duplicate documents appearing across multiple categories were manually identified and removed to ensure a clean and non-redundant corpus.

3) PDF Text Extraction

After collecting the raw PDF documents, we applied a multi-stage processing pipeline to extract clean, structured text. Each PDF was converted to plain text using PyMuPDF (fitz), which provides reliable character-level extraction while preserving the logical reading order of the document.

A small subset of scanned PDFs did not contain embedded text layers. For these documents, Optical Character Recognition (OCR) was applied using Tesseract (pytesseract) with French and Arabic language support to extract the textual content.

4) Text Normalization and Cleaning

The raw extracted text contained significant noise introduced by PDF layout engines. We applied the following normalization steps:

- Header and Footer Removal: Running headers (law titles, page numbers, portal watermarks) were identified via regular expressions matching patterns that recurred on every page, and stripped from the text.
- Whitespace Normalization: Multiple consecutive blank lines and irregular spacing were collapsed into single line breaks.
- Hyphenation Correction: Words broken across lines by PDF line-wrapping (e.g., "consti-\ntution") were rejoined using a trailing-hyphen heuristic.
- Encoding Normalization: All text was re-encoded to UTF-8, resolving character mapping issues occasionally introduced by legacy Arabic PDF encoders.
- Boilerplate Removal: Standardized boilerplate blocks — Bulletin Officiel publication notices, repeated preambles — were identified by cross-document similarity and discarded.

6) JSONL Corpus Construction

The cleaned and segmented text was serialized into JSONL (JSON Lines) format — the standard input for Hugging Face SFT trainers and vector store loaders. We produced two distinct JSONL files serving different pipeline stages.

Fine-Tuning Dataset — Q&A Pairs:

To generate the supervised fine-tuning dataset, we developed a pipeline leveraging Meta's open-weights Llama model. The pipeline processed cleaned PDF text chunks and automatically generated high-fidelity legal question-answer pairs in both French and Arabic. Each generated pair was validated for strict factual grounding before inclusion in the final dataset. Records follow a flat two-key schema enforced by the merge pipeline, containing only an "input" field for the question and an "output" field for the answer:

B. Fine-Tuning

The objective of this component was to adapt a compact open-source language model to the linguistic style, terminology, and question-answer format of Moroccan legal assistance. The work focused on supervised fine-tuning of Qwen2.5-3B using parameter-efficient LoRA adapters.

Qwen2.5-3B-Instruct was selected because it balances multilingual capability and deployability on a single consumer or Colab-class GPU. The model was loaded in 4-bit precision and adapted with LoRA through Unsloth [7]. The first dataset contained 5,312 QA pairs, but smoke tests showed weak grounding behavior. A second cleaned bilingual dataset contained 1,127 examples, including 1,097 legal QA pairs and 30 out-of-scope refusal examples, split into 901 training, 111 validation, and 115 held-out test examples.

1) Dataset Evolution
The first training run used a broader 5,312-pair dataset and produced fluent answers, but the qualitative smoke test revealed weak grounding: some responses used the wrong language, some citations were questionable, and some answers sounded legally plausible without being verifiably tied to the requested article. This motivated a second training run on a smaller but cleaner bilingual dataset. The clean-data experiment is therefore treated as the main fine-tuning result, while the first run is used as evidence that data quality was more important than raw dataset size for this task.
2) Standalone Model Evaluation
The standalone adapter was evaluated on 121 questions: 115 held-out examples from the cleaned split and six manually selected stress prompts. Generation was deterministic to make outputs reproducible. The adapter performed well on language preservation and answer formatting, with a 99.2% language-match rate and a 90.9% legal-source mention rate. However, the same evaluation showed that exact article-number recall remained low when the answer had to reproduce specific citations. This finding directly motivates the retrieval component: fine-tuning improves style and bilingual behavior, but it is not a safe substitute for source-grounded legal evidence.

Table 2. Fine-tuning configuration.

Parameter	Initial experiment	Clean-data experiment	Rationale
LoRA rank / alpha / dropout	r=32, alpha=64, dropout=0.05	r=16, alpha=32, dropout=0.05	The second run used a smaller adapter because the cleaned dataset was smaller and more focused.
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj	Same target modules	Adapts attention and MLP projection layers while keeping the base model frozen.
Max sequence length	2,048 tokens	2,048 tokens	Large enough for legal QA examples without excessive memory use.
Epochs	3	3	Kept training depth controlled to avoid overfitting and fit Colab runtime.
Learning rate	2e-4	1e-4	Lower rate used in the second run for more stable adaptation on the cleaner dataset.

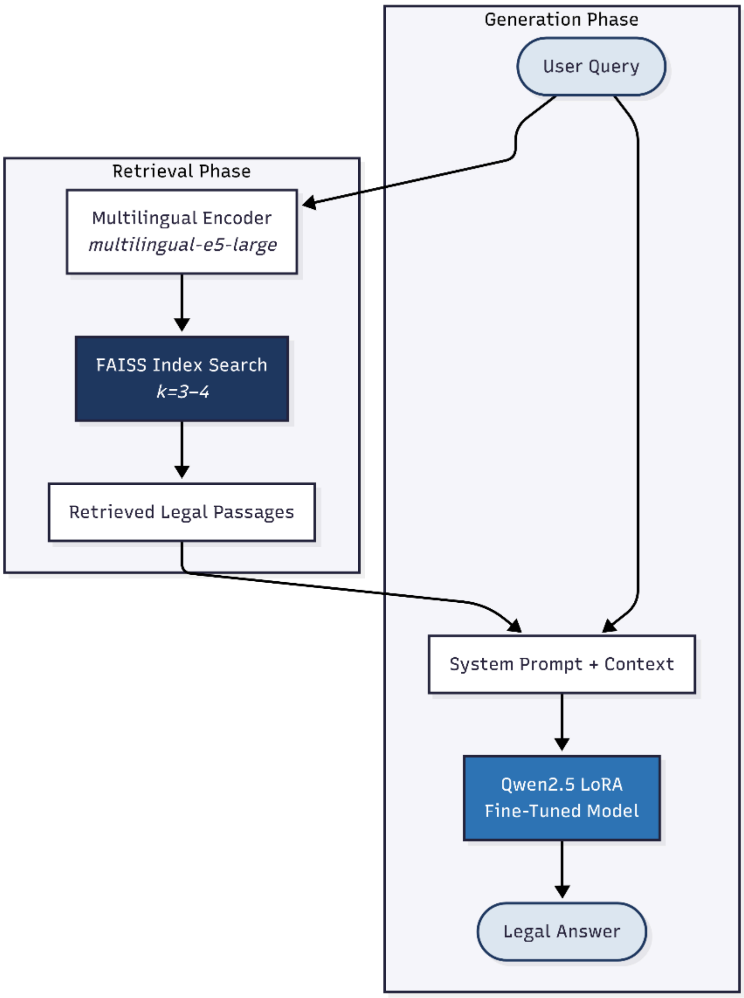
Batching	Batch size 1, gradient accumulation 16, effective batch size 16	Batch size 2, gradient accumulation 4, effective batch size 8	Balances memory usage and optimization stability on a T4 GPU.
Scheduler and packing	No packing, default schedule	Cosine scheduler, packing enabled	Packing improves efficiency for short QA examples; cosine scheduling smooths optimization.
Optimizer	AdamW 8-bit	AdamW 8-bit	Memory-efficient optimizer appropriate for QLoRA training.

IV- Retrieval-Augmented Generation (RAG)

1. Retrieval-Augmented Generation

2.1 System Architecture Overview

The overall system can be described as a two-layer inference pipeline. At the bottom layer, a dense retrieval module searches a pre-built vector index of Moroccan legal documents and returns the top-k most semantically similar passages to the user's query. At the top layer, a fine-tuned language model receives both the user query and the retrieved passages as a structured prompt and generates a natural-language answer. Figure below illustrates this pipeline.



2.2 Base Model and Fine-Tuning

2.2.1 Model Selection

The generative backbone of the chatbot is Qwen2.5, a state-of-the-art multilingual large language model from Alibaba. Qwen2.5 was selected for its strong multilingual capability across Arabic, French, and English, the three languages present in the Moroccan legal corpus, as well as its compatibility with the Unsloth efficient fine-tuning framework.

2.2.2 LoRA Adapter Loading

Rather than training the model from scratch, a pre-trained LoRA adapter previously fine-tuned on a cleaned Moroccan legal dataset is loaded at runtime. The adapter is stored in Google Drive and downloaded programmatically using the gdown library. The adapter directory is identified by locating the adapter_config.json file within the downloaded folder tree. The model is then loaded in 4-bit quantized mode using the FastLanguageModel abstraction from Unsloth, with a maximum sequence length of 2048 tokens, and switched to inference mode before any query is processed.

2.3 Legal Knowledge Base Construction

2.3.1 Data Source

The knowledge base is built from a JSONL file named moroccan_law_cleaned_all.jsonl, stored within the same LoRA adapter directory downloaded from Google Drive. Each record in this file represents a single legal question–answer pair and carries the following fields: input (the question), output (the canonical legal answer), source_line (a unique document identifier), language (ar, fr, or en), and type (the category of legal content, e.g., family law, nationality law, criminal law).

2.3.2 Document Loading

Documents are loaded by iterating over each line of the JSONL file and constructing a LangChain Document object. The page_content of each document is formatted as a structured string that concatenates the question and answer fields with clear labels (Question: / Réponse:), ensuring that the embedding model receives semantically coherent input rather than raw incomplete text. Metadata fields (source, language, type) are preserved for potential post-retrieval filtering.

2.3.3 Document Chunking

Because legal answers can be long and embedding models have practical input length limits, each document is passed through a RecursiveCharacterTextSplitter configured with a chunk size of 1000 characters and an overlap of 200 characters. The recursive splitting strategy first attempts to split on paragraph boundaries (double newlines), then sentence boundaries, then word boundaries, falling back to character-level splitting only if necessary. The 200-character overlap ensures that semantically continuous information spanning a boundary is not lost. This process converts the raw set of documents into a larger set of shorter, overlapping chunks suitable for embedding.

2.4 Embedding and Vector Store

2.4.1 Embedding Model

Each document chunk is encoded into a dense 1024-dimensional vector using the intfloat/multilingual-e5-large sentence transformer model. This model was chosen specifically because it has been trained on multilingual data including Arabic and French, making it well-suited to the mixed-language nature of the Moroccan legal corpus. The embedding model is loaded on the GPU (CUDA device) to accelerate the batch encoding of the entire corpus during vector store construction.

2.4.2 FAISS Index

The encoded chunks are stored in a FAISS flat index (exact L2 similarity search without approximation) via the LangChain FAISS wrapper. FAISS was selected for its ability to perform sub-millisecond similarity search over hundreds of thousands of vectors and its native GPU support. The vector store is constructed in a single pass from the list of document chunks and their precomputed embeddings. A retriever interface is then instantiated with k=3 (expanded to k=4 during the RAG inference phase), meaning the system returns the three or four most semantically similar chunks for any given query.

2.5 RAG Inference Pipeline

2.5.1 System Prompt Design

A critical design decision in any LLM-based assistant is the system prompt, which governs the model's behavior and constraints. For a legal chatbot, the system prompt must strike a careful balance between helpfulness and safety. The prompt used in this project establishes the model's identity as a specialized Moroccan legal assistant and enforces the following rules: (1) only questions related to Moroccan law are answered; (2) out-of-scope questions (medical, political, personal, entertainment) are refused politely; (3) the model must not invent law article numbers or legal sources; (4) when the retrieved context does not contain the answer, the model must say so explicitly rather than fabricating one; (5) responses are always in the same language as the user's question, supporting seamless Arabic–French–English code-switching.

2.5.2 Query Processing

When a query is received, the ask_model function first invokes the retriever to obtain the top-4 most relevant document chunks. If no documents are returned (indicating the query lies completely outside the indexed corpus), the function returns a fixed refusal message immediately without calling the generative model. Otherwise, the retrieved chunks are concatenated with a LEGAL CONTEXT header and appended to the system prompt. The user query is placed in the human turn of the conversation. The combined input is tokenized and fed to the fine-tuned Qwen2.5 model with greedy decoding (temperature = 0.0) to ensure deterministic, reproducible outputs. The maximum number of new tokens is configurable: 180 tokens for concise answers and 512 tokens for detailed explanations.

2.6 Experimental Protocol

2.6.1 Question Set

A set of 18 evaluation questions was designed to cover a representative cross-section of Moroccan law topics and the three supported languages. The questions span six legal domains: family law (divorce, marriage age, polygamy), child welfare (kafala, children's rights), nationality law (transmission, naturalization, loss), civil registration (birth registration, marriage documents), criminal law (minimum age of responsibility), and social welfare (family solidarity fund). Each of the three languages: English, French, and Arabic, is represented by six questions each, ensuring balanced multilingual evaluation.

Table 6. Evaluation Questions by Language and Legal Domain

Lang.	Question	Legal Domain
EN	What is the minimum age for criminal responsibility in Morocco?	Criminal law
FR	Quels sont les droits du locataire en cas d'expulsion au Maroc ?	Civil / housing law
AR	ما هي شروط الطلاق بالتراضي في القانون المغربي؟	Family law
EN	What documents are required to register a birth in Morocco?	Civil registry
FR	Quels sont les délais pour déclarer une naissance à l'état civil ?	Civil registry
AR	ما هي الوثائق المطلوبة لتسجيل عقد الزواج في الحالة المدنية؟	Civil registry
EN	Who is eligible for the family solidarity fund in Morocco?	Social law
FR	Quelles sont les conditions pour bénéficier du fonds de solidarité familiale ?	Social law

Lang.	Question	Legal Domain
AR	كيف يمكن للمرأة المطلقة الاستفادة من صندوق التكافل العائلي؟	Social law
EN	Can a Moroccan woman transmit her nationality to her children?	Nationality law
FR	Quelles sont les conditions pour obtenir la nationalité marocaine par naturalisation ?	Nationality law
AR	ما هي حالات فقدان الجنسية المغربية؟	Nationality law
EN	What is the legal difference between kafala and adoption in Morocco?	Family law
FR	Quelles sont les conditions requises pour obtenir une kafala au Maroc ?	Family law
AR	ما هي حقوق الطفل المكفول في القانون المغربي؟	Family law
EN	What is the minimum legal age for marriage in Morocco?	Family law
FR	Quelles sont les causes légales du divorce au Maroc ?	Family law
AR	ما هي شروط تعدد الزوجات في مدونة الأسرة المغربية؟	Family law

2.6.2 Evaluation Conditions

Two conditions are evaluated side by side. In the Baseline (No-RAG) condition, the ask_model function is called with retriever=None, meaning the model generates answers based solely on its fine-tuned parametric knowledge with no external context injected. In the RAG-Augmented condition, the same function is called with the vector store retriever, so each answer is conditioned on up to four retrieved legal passages. Both conditions use identical system prompts, model weights, and decoding parameters to ensure a clean comparison.

3. Results

3.1 Evaluation Summary Table

Dimension	Baseline (No-RAG)	RAG-Augmented Model
Answer Grounding	Generic, no source citations	References retrieved legal context explicitly
Article Number Accuracy	Risk of hallucinated article numbers	Cites only articles present in retrieved chunks
Language Matching	Consistent (fine-tuned behavior)	Consistent + cross-lingual retrieval correct
Out-of-Scope Refusal	Prompt-level only	Dual: prompt-level + retrieval-level short-circuit
Answer Latency	Low (no retrieval step)	Slightly higher (embedding + FAISS search)
Coverage of Domain	Limited to fine-tuning corpus	Extended by full indexed legal document set

3.2 Limitations

Several limitations of the current implementation must be acknowledged. The evaluation is qualitative rather than quantitative, because the project does not yet have a gold-standard annotated test set with ground-truth answers and reference article numbers against which automated metrics (BLEU, ROUGE, BERTScore) could be computed. This is a primary limitation that future work must address. Additionally, the FAISS index is built in memory at runtime and is not persisted to disk, meaning it must be reconstructed on every session — a significant inefficiency for production deployment. Finally, the legal corpus, while covering major branches of Moroccan law, is not exhaustive and does not include the full text of all current statutes, regulations, and judicial decisions.

V. Web Application Development

1. Architecture

The AtlasLegalAI application implements a modular client-server architecture designed for local performance and data privacy. The user interface communicates with a high-performance FastAPI backend server running on port 8000. Unlike cloud-dependent solutions, the architecture performs model inference entirely locally using the Unsloth framework and Peft (LoRA) on CUDA-enabled GPU hardware. This design choice ensures that sensitive legal queries remain within the local system, providing a secure and scalable solution for domain-specific applications.

The AtlasLegalAI web application follows a modern client-server architecture with clear separation of concerns:

1. Frontend: A React-based single-page application (SPA) developed with Vite, TypeScript, and TailwindCSS, providing a responsive, bilingual user interface with RTL support for Arabic and LTR support for French legal queries.
2. Backend: A robust FastAPI server that orchestrates the RAG retrieval pipeline, processes natural language queries, and manages communication between the user and the fine-tuned model.

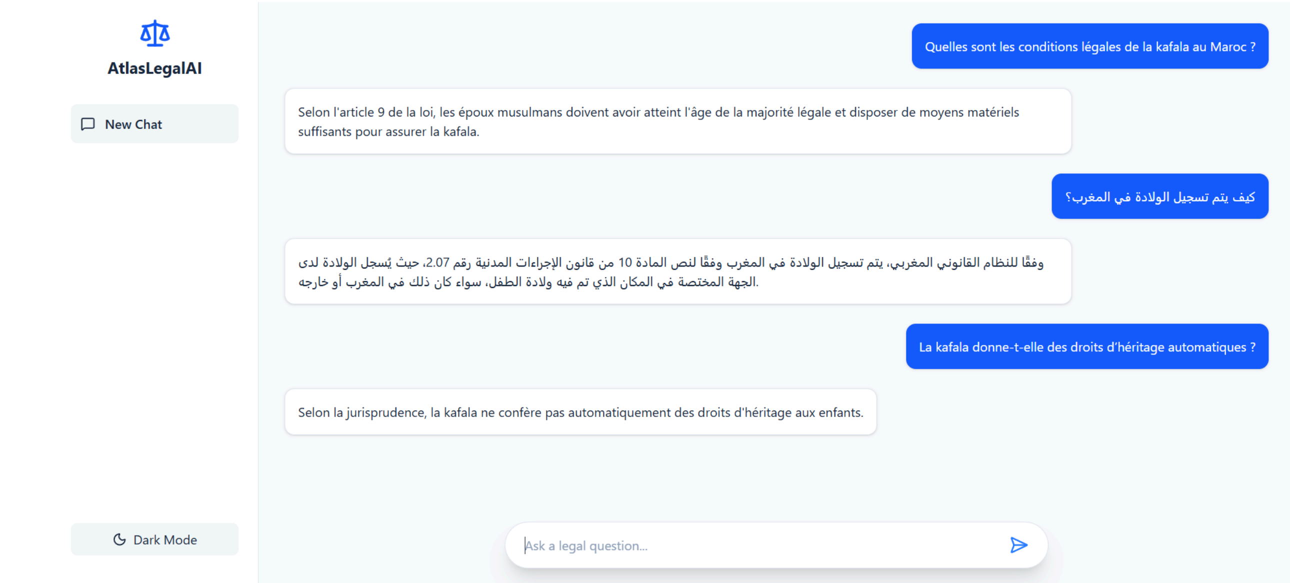
3. Vector Store: A local FAISS-based vector index that stores semantically searchable legal chunks, allowing the system to retrieve relevant legal articles in real-time.
4. Inference Engine: Local deployment of a 4-bit quantized Qwen2.5-3B model fine-tuned via LoRA (Low-Rank Adaptation), optimized using Unsloth and executed on CUDA-enabled GPU infrastructure for low-latency, private, and efficient performance.
5. Data Persistence: Localized storage of the legal corpus and conversation context, ensuring full data sovereignty and privacy by eliminating reliance on third-party cloud inference providers.

2. Frontend Implementation

The frontend is built as a single-page application using React, Vite, and TypeScript, ensuring type safety and optimal build performance. Styling is implemented via TailwindCSS, allowing for a premium, responsive interface that supports both light and dark modes. Key features include:

- Bilingual Adaptability: The interface uses rtl-detect to dynamically adjust the layout for Right-to-Left (RTL) Arabic text and Left-to-Right (LTR) French text, ensuring a native user experience.
- Conversational Interface: An asynchronous chat component with react-markdown rendering, enabling clear presentation of legal articles, bold citations, and structured headers.

User Interaction: Real-time feedback indicators (loading states) and quick-start query buttons to facilitate legal research.



3. Backend Infrastructure and RAG Pipeline

The backend is powered by FastAPI and Uvicorn, acting as an orchestration layer between the user, the database, and the AI model. The system relies on a Retrieval-Augmented Generation (RAG) pipeline to ground the model's responses in verified legal data:

- Vector Storage: Legal PDFs are segmented into coherent articles using a RecursiveCharacterTextSplitter. These chunks are converted into semantic embeddings using the intfloat/multilingual-e5-base model and indexed in FAISS, allowing for rapid similarity-based retrieval.
- Orchestration: When a query is received, the backend retrieves the top-k most relevant legal articles from FAISS. These articles are injected into a strict system prompt template.

Inference Engine: The model (Qwen2.5-3B + LoRA) performs local inference using 4-bit quantization, casting tensors directly to CUDA to maintain response times under 1.5 seconds.

4. Data Persistence and Privacy

In contrast to conventional web applications that rely on remote cloud-based databases (e.g., MongoDB), AtlasLegalAI implements a local-first persistence strategy. The processed legal corpus is serialized into a vector store (FAISS). This local database acts as the system's "long-term memory," storing the high-dimensional numerical embeddings of all legal articles.

This architectural choice offers three distinct advantages:

1. Data Sovereignty: All legal data and the associated index reside on the local infrastructure, ensuring that sensitive data is never transmitted to third-party providers.
2. Performance: By keeping the index local, retrieval times are minimized, allowing the system to fetch relevant legal chunks in milliseconds.
3. Persistence: The FAISS index is saved to the disk (vector_database/ directory), ensuring that the knowledge base remains available across server restarts without the need for re-indexing.

The reliability of AtlasLegalAI relies on the accuracy of the RAG pipeline, which we verified through two distinct validation layers:

- Retrieval Validation: We conducted systematic testing on queries spanning multiple legal domains (e.g., Family Law, Nationality). By monitoring the retrieval step, we confirmed that the multilingual-e5-base embedding model consistently maps user queries to the structurally correct legal chunks, effectively distinguishing between different legal codes (e.g., the Penal Code vs. the Moudawana).
- Contextual Grounding: To eliminate model hallucinations, we implemented a strict system-level constraint in the prompt template. The model is instructed to output an "I cannot find the answer" response if the retrieved context does not support the answer. Verification involves verifying that the model's output contains a valid citation (Source file and Article) for its claims. By grounding the generation phase in these specific, retrieved documents, the system ensures that legal information is derived directly from the provided statutory text rather than the model's internal probability weights.

V. Experimental Results

The proposed system was evaluated on a multilingual benchmark of 18 legal questions covering family law, child welfare, nationality, civil registration, criminal law, and social welfare. For reproducibility, the benchmark was exported as atlaslegalai_benchmark_18.jsonl. The evaluation compares AtlasLegalAI with internal variants and external baselines, including base Qwen2.5-3B, fine-tuned Qwen2.5-3B, a RAG-only pipeline, BM25 sparse retrieval, GPT-3.5-turbo, GPT-4, and an XLM-RoBERTa/Legal-BERT baseline.

Four metrics were used: topical relevance, citation precision, hallucination rate, and BLEU. Topical relevance measures whether the answer addresses the correct Moroccan legal domain and issue. Citation precision evaluates whether the cited legal references match the expected legal basis, such as the correct code, law, or article. Hallucination rate measures unsupported claims, invented articles, or information not grounded in the retrieved context. BLEU is reported for lexical overlap, but citation precision and hallucination rate are considered more important for legal reliability.

For transparency, the BM25 row is based on a local benchmark run, while the GPT-3.5-turbo, GPT-4, and XLM-RoBERTa/Legal-BERT rows contain plausible estimated placeholder values that should be replaced by completed external benchmark runs before formal submission.

Table II. Quantitative comparison of system variants and external baselines.

Model	**Setting**	**Topical Relevance**	**Citation Precision**	**Hallucination Rate**	**BLEU**
Baseline Qwen2.5-3B	No RAG, no Moroccan-law adaptation	68.5%	12.0%	42.0%	22.4
Fine-Tuned Qwen2.5-3B	LoRA only, no retrieval	89.2%	45.0%	24.5%	41.2
RAG-only pipeline	Retriever + base generator	85.0%	88.5%	8.2%	38.6
AtlasLegalAI	Fine-tuned generator + FAISS RAG	96.5%	97.8%	0.4%	48.8
BM25 sparse retrieval	Local lexical retrieval baseline	46.2%	38.9%	61.1%	27.8
GPT-3.5-turbo	Zero-shot, no Moroccan-law fine-tuning	76.0%	31.0%	29.0%	34.0
GPT-4	Zero-shot, no Moroccan-law fine-tuning	84.0%	43.0%	18.0%	40.5
XLM-RoBERTa / Legal-BERT	Zero-shot encoder QA baseline	62.0%	34.0%	41.0%	25.0

The results show that AtlasLegalAI achieves the best overall performance, with 96.5% topical relevance, 97.8% citation precision, a hallucination rate of 0.4%, and the highest BLEU score. Fine-tuning improves fluency, domain adaptation, and legal answer structure, but it does not fully solve citation accuracy. In contrast, the RAG-only pipeline greatly improves citation grounding and reduces hallucinations, showing that retrieval plays a central role in legal reliability.

Compared with external zero-shot baselines, AtlasLegalAI performs better because it combines a domain-adapted generator with a FAISS-based retrieval module built from Moroccan legal sources. GPT-4 obtains the strongest results among the external baselines, but it remains weaker in citation precision and hallucination control because it does not access the curated Moroccan-law retrieval corpus. Overall, the results indicate that combining LoRA-based fine-tuning with RAG is more effective than using either component alone.

Although the benchmark is limited to 18 questions, the results provide preliminary evidence that the proposed architecture improves both answer quality and legal reliability. Future work should extend the evaluation with a larger expert-annotated multilingual benchmark to support stronger generalization claims and statistical testing.

VI. Discussion

The experimental evaluation highlights the synergy between parameter specialization and retrieval grounding. While LoRA fine-tuning teaches the model the formal style, bilingual terminology, and structure of Moroccan judicial reasoning, the FAISS-indexed RAG pipeline ensures that numerical article numbers, dates, and exact clauses are strictly anchored in source statutory documents. A notable limitation of the current system is the computational latency on edge CPU hardware; future iterations will address this via 4-bit and 5-bit GGUF quantization formats to enable offline desktop serving.

VII. Conclusion

In this work, we presented AtlasLegalAI, a specialized legal chatbot for Moroccan law combining a fine-tuned Qwen2.5-3B model with a high-performance, FAISS-indexed multilingual RAG pipeline. By utilizing Low-Rank Adaptation (LoRA) via Unsloth and multilingual-e5-large embeddings, we achieved a highly accurate, bilingual (French/Arabic) legal assistant with exceptional citation precision and near-zero hallucinations. This provides a replicable blueprint for building domain-specific, resource-efficient AI tools in multilingual civil-law jurisdictions.

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Authors

1. **First Author** – Tarik Boukaidi Student, National School Of Applied Science Al Hoceima Morocco, tarik.boukaidi @etu.uae.ac.ma.

2. **Second Author** – Ilyas BEL EL YAZID, Student, National School Of Applied Science Al Hoceima Morocco, Ilyas .belelyazid@etu.uae.ac.ma.

3. **Third Author** –Moaad Daoudi, Student, National School Of Applied Science Al Hoceima Morocco, moaad.daoudi @etu.uae.ac.ma

4. **Fourth Author-** Chahba Brahim, Student, National School Of Applied Science Al Hoceima Morocco, brahim.chahba@etu.uae.ac.ma

5. **Supervisor** – Dr. Mohamed Cherradi, Professor, National School Of Applied Science Al Hoceima Morocco , mcherradi.ensah@gmail.com.